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This document presents the architecture for **Originator-Led Automated Trade Finance**.
In this model, the Originator acts as the "Master Orchestrator," managing all real-world legalities and document verification, while the Smart Contract executes the **Delivery-vs-Payment (DvP)** state changes to ensure a trustless environment for Liquidity Providers (LPs).

1. THE TOKEN ARCHITECTURE: THE "BASKET"

The Originator digitizes the trade by minting a multi-layered NFT structure. This creates a standardized "unit of credit" that is legally enforceable and DeFi-ready.

- **invoiceNFT (The Context):** Digital evidence of the commercial transaction (line items, shipping data).
- **promNFT (The Obligation):** The **Promissory Note**. A legally binding, negotiable instrument signed by the Buyer promising to pay the bearer.
- **titleNFT (The Collateral):** The **Bill of Lading**. Represents legal ownership of the physical goods.
- **boeNFT (The Master Wrapper):** The **Bill of Exchange**. A single token that "wraps" the above three. This is the asset the Vault holds.

2. ORIGINATOR-LED WORKFLOW (Ready-to-Ship)

The Originator manages the interaction between Buyer and Seller, ensuring the state machine only progresses when physical and digital conditions are met.

Phase 1: Preparation (The "Commitment")

- **Originator Action:** Verifies the goods are "Ready to Ship" and the Buyer's credit is active.
- **The Mint:** Originator mints the boeNFT (containing Invoice, PromNote, and Title).
- **State Change:** The titleNFT is moved into **Smart Contract Escrow**. The Seller retains physical possession in a bonded warehouse or port but loses legal title.

Phase 2: Funding (The "Atomic Swap")

- **Originator Action:** Triggers the funding request to the **Credit Vault**.
- **The Swap:** The Vault deposits **USDC**. The contract simultaneously:
    1. Sends **USDC** to the **Seller** (Immediate liquidity).
    2. Sends the **boeNFT** to the **Vault/LP**.
- **State Change:** Seller is paid. The **Vault** now legally owns the debt and the physical collateral.

Phase 3: Maturity & Settlement (The "Release")

- **Originator Action:** Notifies the Buyer of the payment due date.
- **Repayment:** Buyer deposits **USDC** into the Smart Contract.
- **The Great Release:**
    1. **USDC** \rightarrow Sent to the **Vault/LPs** (Principal + Interest).
    2. **titleNFT** \rightarrow Sent to the **Buyer** (Legal title is released + can claim goods).

3. **boeNFT** \rightarrow **Burned/Extinguished**.

3. SAFEGUARDS & SECONDARY MARKET LOGIC

A. Title Retention (The Safeguard)

The **titleNFT** is the ultimate lock. The Buyer cannot legally resell or clear the goods from customs without it. The contract ensures the title only moves when the USDC settlement is 100\% verified.

B. Secondary Market & Accounts Receivable (AR)

Because the boeNFT is a standardized negotiable instrument, the Originator can facilitate a **Secondary Market**:

- **LP Exit:** An original LP can sell their boeNFT to another investor mid-transit. The legal claim (promNFT) and the collateral (titleNFT) move automatically to the new owner.
- **Fractionalization:** Large trade assets can be split into fungible tokens, allowing a "Basket of Credit" to be shared among thousands of retail LPs.

C. Originator as "Technical Bailee"

If a Buyer defaults, the Originator—acting on behalf of the LP—uses the **titleNFT** to reclaim the physical goods and the **promNFT** to pursue legal debt collection.

4. FINAL STATE TRANSITION SUMMARY

Trigger Event	Asset (Physical)	Title (Legal)	Payment (USDC)
Ready to Ship	At Dock	Held by Seller	Idle in Vault
Commitment	In Transit	**Locked in Escrow**	Idle in Vault
Funding Swap	In Transit	**Owned by Vault**	**Sent to Seller**
Settlement	**Arrived**	**Owned by Buyer**	**Sent to LP/Vault**

Summary for Vault Managers: This architecture moves TradeFi from manual "Document Processing" to automated "State Management." By having the **Originator** manage the Buyer/Seller relationship and the **Smart Contract** manage the DvP settlement, you create a scalable, senior-secured DeFi fund backed by real-world movement of goods.